Tarun Naithani

Computational Statistics · Data Science · Machine Learning

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WORK EXPERIENCE

Senior Data Scientist
07/2025 – Present

Sanofi — Hyderabad, India

- Developing scalable deep learning pipelines using **PyTorch**, **PyTorch Geometric (PyG)**, and **distributed training frameworks** for large-scale model training and graph-based computation.
- Spreading the implementation of **Graph Neural Network (GNN)** architectures for **clinical trial outcome prediction** and **adverse event forecasting**, integrating structured biomedical data.
- Developed **LLM-powered tools** for automated curation and structuring of clinical trial data, enabling seamless enrichment of downstream knowledge graphs (AWS Bedrock).
- Implemented **Variational Autoencoders (VAE)** on **proteomics and multi-omics datasets** to model latent representations for **drug response prediction**.
- Contributed to **equivariant neural network** projects for **molecular property prediction** (e.g., solubility), incorporating geometric deep learning principles.
- Completed internal **immunology training** (e.g. T-cell biology, immune signalling, immune responses) to bridge biological context with computational and AI-based analysis.

Data Scientist
01/2022 – 12/2024

Oxford Biodynamics — Oxford, United Kingdom

- Developed modular, production-ready Python code adhering to PEP8 standards, supporting in-house engineering and research workflows.
- Enhanced epigenetic interaction discovery by implementing **GraphSAGE and GCN** deep learning models, improving genomic data representation and link prediction accuracy (PyTorch Geometric).
- Built an application around **knowledge graphs**, leveraging graph machine learning for genomic insights and implementing graph completion algorithms for knowledge extraction (PyTorch, MongoDB, SQL, Shiny, Dash).
- Constructed **command-line interfaces (CLI)** using Python for streamlined execution of graph-based analytics workflows and ML model evaluation.
- Packaged reusable **Python and R libraries** for internal teams and supported **Optuna**-based hyperparameter tuning for **XGBoost** and **CatBoost** models.
- Used **Docker** and **Google Cloud Platform (GCP)**, enabling scalable containerized tools for cross-team collaboration.
- Created and maintained statistical models (**GLM**, **GAM**) and applied **causal inference** with **Bayesian networks** for understanding biomarker relationships.
- Contributed to **development workflows** using Git, ensuring rapid prototyping and collaborative iteration.
- Communicated insights and results to stakeholders via data storytelling, visualizations, and presentations to guide diagnostics R&D.

Quantitative Research (Intern)
12/2019 – 04/2020

Taras Capital, Hooklabs GmbH

- Designed machine learning models to detect trading signals and patterns in time-series financial data using Python and Scikit-learn.
- Built real-time data pipelines for ingesting tick and order book data using **AWS**, streamlining the backtesting of algorithmic strategies.

SKILLS

Python | R | NumPy/JAX | Pandas | SQL/MongoDB | PyTorch/PyG | Git/version control | Matplotlib/Seaborn | Shiny/Dash/Streamlit | Docker | Deep learning | Argparse (CLI)

EDUCATION

M.Sc in Statistics
University of Glasgow
01/2021 – 01/2022

B.Tech in Marine Engineering
Indian Maritime University
08/2014 – 04/2019

PUBLICATIONS

Cancers

Circulating Chromosome Conformation Signatures Significantly Enhance PSA Positive Predicting Value and Overall Accuracy for Prostate Cancer Detection
29 January 2023 · Contributed

Annals of the Rheumatic Diseases

OP0257 A Peripheral Blood Chromosome Conformation Signature Can Predict Flare Following DMARD Cessation in Rheumatoid Arthritis
2023 · Contributed

- Participated in research and documentation for the development of automated trading systems; contributed to model validation and strategy tuning.

Assistant Engineer Officer 04/2018 – 12/2018
Essar Shipping Ltd, International

- Oversaw technical operations and safety compliance of marine vessels, ensuring optimal mechanical performance and minimizing downtime.
- Coordinated preventive maintenance and troubleshooting of onboard systems, gaining experience in high-stakes operational environments.

PERSONAL PROJECTS

regulatr — Cell-type-aware non-coding variant prioritization tool

- Built as part of **Built with Claude: Life Science Hackathon**, with a demo.

CLI RAG tool with LangChain & Pinecone (prototype stages) 04/2025

- Tutorial on building a vector database and CLI tool to enable semantic search and information retrieval from unstructured scientific text.

Python Package – Drug-Disease Recommender (LightGCN) 12/2024

- Designed and developed a Python package to recommend drug-disease pairs using **Light Graph Convolutional Networks (LightGCN)**.
- Achieved over **0.80 AUC** on validation datasets and packaged the tool with documentation and modular components for reuse.

Time Series to Visibility Graph Python Package 02/2025

- Built a tool to encode time series as **visibility graphs**, capturing spatial-temporal dependencies in sequential data.

Bayesian Predictive Interval Model for Team Wins (Bayesian Stats) 11/2024

- Developed a Bayesian regression model using **RStan** to predict team win probabilities based on prior season performance and betting odds.
- Incorporated Gaussian priors on coefficients and built functions to calculate **predictive intervals** for match outcomes.

Graph Classification for Molecular Properties (Experimental approach) 10/2024

- Achieved **0.75 validation accuracy** using PyTorch Geometric, with optimized training routines and visual tracking of model performance. GIN architecture.

Dissertation – Classification Analysis for ARDS Patients 09/2021 – 12/2021

- Analyzed patient biomarkers to classify outcomes in Acute Respiratory Distress Syndrome (ARDS), including ECMO response analysis.

Cancers

Development and Validation of Blood-Based Predictive Biomarkers for Response to PD-1/PD-L1 Checkpoint Inhibitors: Evidence of a Universal Systemic Core of 3D Immunogenetic Profiling across Multiple Oncological Indications
2023 · Contributed

INTERESTS

Math and Computing topics
AI for healthcare · Swimming
Reading · Music · Philosophy